

SYED MUHAMMAD RAFAY ALI

Full Stack Developer | Machine Learning Engineer

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SUMMARY

Computer Science graduate (2022–2026) from FAST-NUCES with hands-on experience building full-stack web applications across the MERN stack and TypeScript. Shipped six end-to-end projects spanning booking platforms, AI-driven event management, and compiler design, with applied exposure to classical machine learning techniques including neural networks and K-Means clustering.

EDUCATION

Bachelor of Science in Computer Science

National University of Computer and Emerging Sciences (FAST-NUCES) | 2022 – 2026

SKILLS

Programming Languages: C/C++, JavaScript, TypeScript, Python

Frontend: React, React Router, HTML5, CSS3, Tailwind CSS, Bootstrap, jQuery, Vite

Backend: Node.js, Express.js, JWT, Nodemon

Databases & Tools: MongoDB, PostgreSQL, Supabase, Git, GitHub, Vercel, Flex, Bison, Make

Languages: English (Fluent), Urdu (Fluent)

PROJECTS

HostelSpot: Student Hostel Booking Platform

Tech Stack: React, TypeScript, Supabase, Tailwind CSS

- Developed a full-stack hostel booking platform with real-time availability updates and dual dashboards for students and hostel owners, plus an admin management panel.
- Set up a CI/CD pipeline using GitHub Actions to automate testing and deployment.

CarInsight Pro: Car Research & Comparison Platform

Tech Stack: React, Node.js, MongoDB, Express

- Built a MERN-stack platform for side-by-side car specification comparisons, featuring an interactive matchmaking quiz and a custom-built CMS for content management.
- Set up a CI/CD pipeline using GitHub Actions to automate testing and deployment.

PakTravel AI System: Intercity Bus Travel System

Tech Stack: Python, scikit-learn, Pandas, NumPy

- Implemented an AI-powered travel system combining classical search algorithms, propositional logic, and CSP-based scheduling; applied neural networks and K-Means clustering for route and demand analysis.

Interactive 3D Portfolio

Tech Stack: React, TypeScript, Vite, Tailwind CSS

- Built a visually rich personal portfolio with a custom mathematical SVG character with cursor-tracking eyes, custom shaders, and scroll-based reveal animations, all without third-party animation libraries.

FestivAI: AI-Powered Event Management Platform

Tech Stack: React, Node.js, Express, MongoDB

- Built an intelligent event-management platform with RSVP-driven booking, dynamic QR check-ins, venue management, AI-generated e-card invitations, and Google Maps API integration for real-time hall locations.

MiniC Compiler

Tech Stack: C, Flex, Bison, GCC/MinGW, Make

- Built a 7-phase compiler in C covering lexical, syntax, and semantic analysis through intermediate code generation, optimization, and target code generation, with constant folding and dead code elimination.
- Designed a TAC interpreter to execute the optimized intermediate code and produce real program output for validation.